

DATA CLEANING PIPELINE

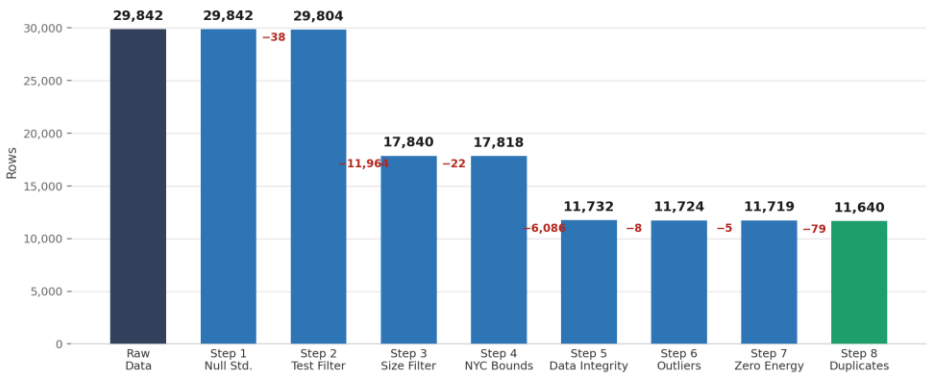
LL97 Data Cleaning & Reduction Report

Row-level summary of the filtering pipeline applied to the raw LL97 property extract, from initial ingest through to the final compliance-ready dataset.

29,842 INITIAL RAW ROWS	11,640 FINAL CLEAN ROWS	18,202 TOTAL ROWS DROPPED	61.0% OVERALL REDUCTION
----------------------------	----------------------------	------------------------------	----------------------------

Row Count by Step

The dataset narrows from 29,842 raw rows to 11,640 final rows across eight sequential filters. The Size Threshold and Data Integrity filters account for the large majority of the reduction.



Cleaning Pipeline Detail

Step	Name	Description	Before	After	Dropped
Step 1	Standardizing Null Values	Replaced the text "Not Available" with standard system NULL (NaN) values to allow numeric filtering. This step doesn't drop rows, but prepares the data.	29,842	29,842	0
Step 2	Filtering Test Properties	Removed properties marked as "Test" in the Construction Status column, keeping only real assets.	29,842	29,804	-38
Step 3	LL97 Size Threshold Filter	Filtered properties to only include buildings with a Gross Floor Area (GFA) ≥ 50,000 sq ft, matching the Local Law 97 regulatory baseline.	29,804	17,840	-11,964
Step 4	New York City Boundaries Filter	Processed typo-ridden city names and filtered the dataset to strictly contain addresses within New York City boroughs (dropping out-of-state/invalid locations).	17,840	17,818	-22
Step 5	Data Integrity & Alerts Filter	Dropped records containing critical missing or overlapping meter data (e.g., energy gaps, less than 12 months of data) to ensure emissions baseline accuracy.	17,818	11,732	-6,086
Step 6	Penalty Outliers Filter	Calculated projected carbon penalties and removed extreme statistical outliers (Penalty > \$1700/sqft) likely resulting from data entry errors.	11,732	11,724	-8
Step 7	Zero Energy Use Filter	Dropped records where reported site energy use was exactly zero (which is physically impossible for operating buildings).	11,724	11,719	-5
Step 8	Remove Duplicate Property IDs	Removed duplicate rows based on Property ID, retaining only unique buildings in the final dataset.	11,719	11,640	-79